

LVM Cheat Sheet

Pool disks together and carve them into volumes you can grow, move and snapshot without repartitioning.

How the layers fit

Disks	/dev/sdb	/dev/sdc	← whole disks or partitions
PVs	[PV]	[PV]	← pvcreate: "LVM may use this disk"
VG	[vg0: one pool of space]		← vgcreate: pool the PVs together
LVs	[data] [logs] [free]		← lvcreate: slices of the pool
Filesystems	ext4	xfs	← mkfs + mount, like any partition

Using it

```
sudo pvcreate /dev/sdb /dev/sdc ← writes an LVM label on each disk
sudo vgcreate vg0 /dev/sdb /dev/sdc ← makes the pool vg0
sudo lvcreate -L 20G -n data vg0 ← a 20 GB volume called data
sudo lvcreate -l 100%FREE -n logs vg0 ← small -l = a percentage: all space left
sudo mkfs.ext4 /dev/vg0/data && sudo mount /dev/vg0/data /srv ← then add to /etc/fstab
```

/dev/vg0/data = /dev/mapper/vg0-data: the same volume (in mapper names a - is doubled: **my--vg**)

Reading pvs, vgs and lvs

```
PV          VG Fmt Attr PSize   PFree ← sudo pvs
/dev/sdb    vg0 lvm2 a--  <500.00g    0 ← this disk is fully used
/dev/sdc    vg0 lvm2 a--  <500.00g <294.99g
VG #PV #LV #SN Attr   VSize   VFree ← sudo vgs
vg0   2   3   1 wz--n- 999.99g <294.99g ← VFree = space left to hand out
LV  VG Attr       LSize   Pool Origin Data%  Meta% ← sudo lvs
data vg0 owi-aos--- 600.00g ← o = has a snapshot · a = active · o = open (mounted)
logs vg0 -wi-ao---- 100.00g ← - = plain volume
snap vg0 swi-a-s---  5.00g    data  12.40 ← snapshot of data, 12% full
```

Growing (online, no unmount)

Command	Does
sudo vgextend vg0 /dev/sdd	Adds a new disk to the pool (makes it a PV for you)
sudo lvextend -r -L +10G vg0/data	Grows data by 10 GB; -r grows the filesystem too (ext4 and xfs)
sudo lvextend -r -l +100%FREE vg0/data	Gives data all the free space in the VG
sudo pvresize /dev/sdb	After enlarging a virtual disk: tells LVM the PV got bigger

Shrinking: xfs can't. ext4 only unmounted: back up, **umount**, then **lvreduce -r -L 50G vg0/data** (never without **-r**)

Snapshots

Command	Does
sudo lvcreate -s -L 5G -n snap vg0/data	Freezes data as it is now; 5 GB holds the changes made after
sudo mount -o ro /dev/vg0/snap /mnt	Look at or back up the frozen copy (xfs: add nouuid)
sudo lvconvert --merge vg0/snap	Rolls data back to the snapshot (if in use: at next activation)
sudo lvremove vg0/snap	Deletes the snapshot when you're done

A snapshot whose **Data%** reaches 100% becomes invalid. They slow writes: don't keep them long.

Handy recipes

Command	Does
sudo pvmove /dev/sdb	Moves all data off sdb to other PVs, while running
sudo vgreduce vg0 /dev/sdb && sudo pvremove /dev/sdb	Then takes the empty disk out of LVM
sudo lvs -o +devices	Which disks each volume lives on
sudo lvremove vg0/logs	Deletes a volume and its data (unmount first)

Remember: **lvextend -r** grows the volume and the filesystem in one step — no unmount, no reboot.